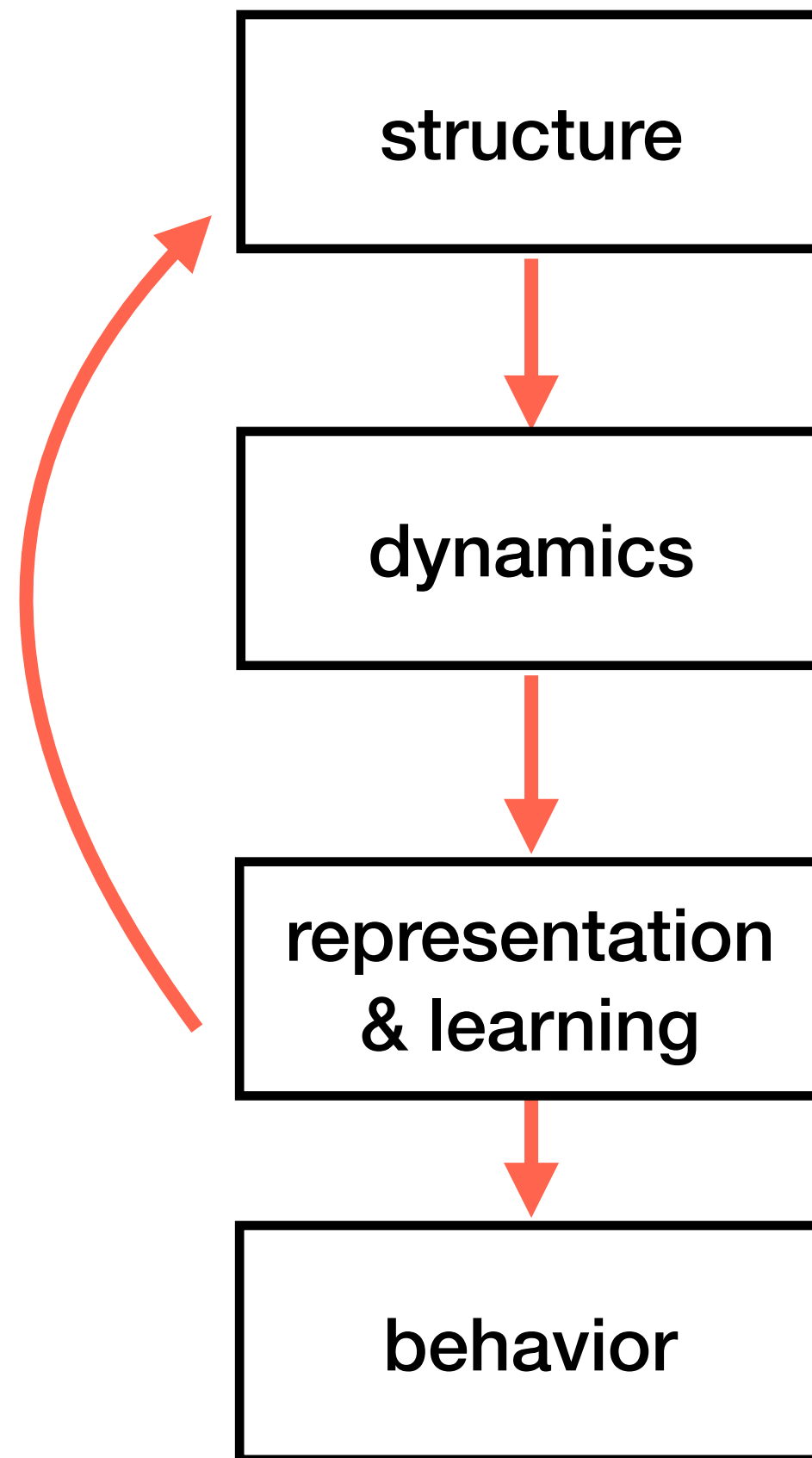
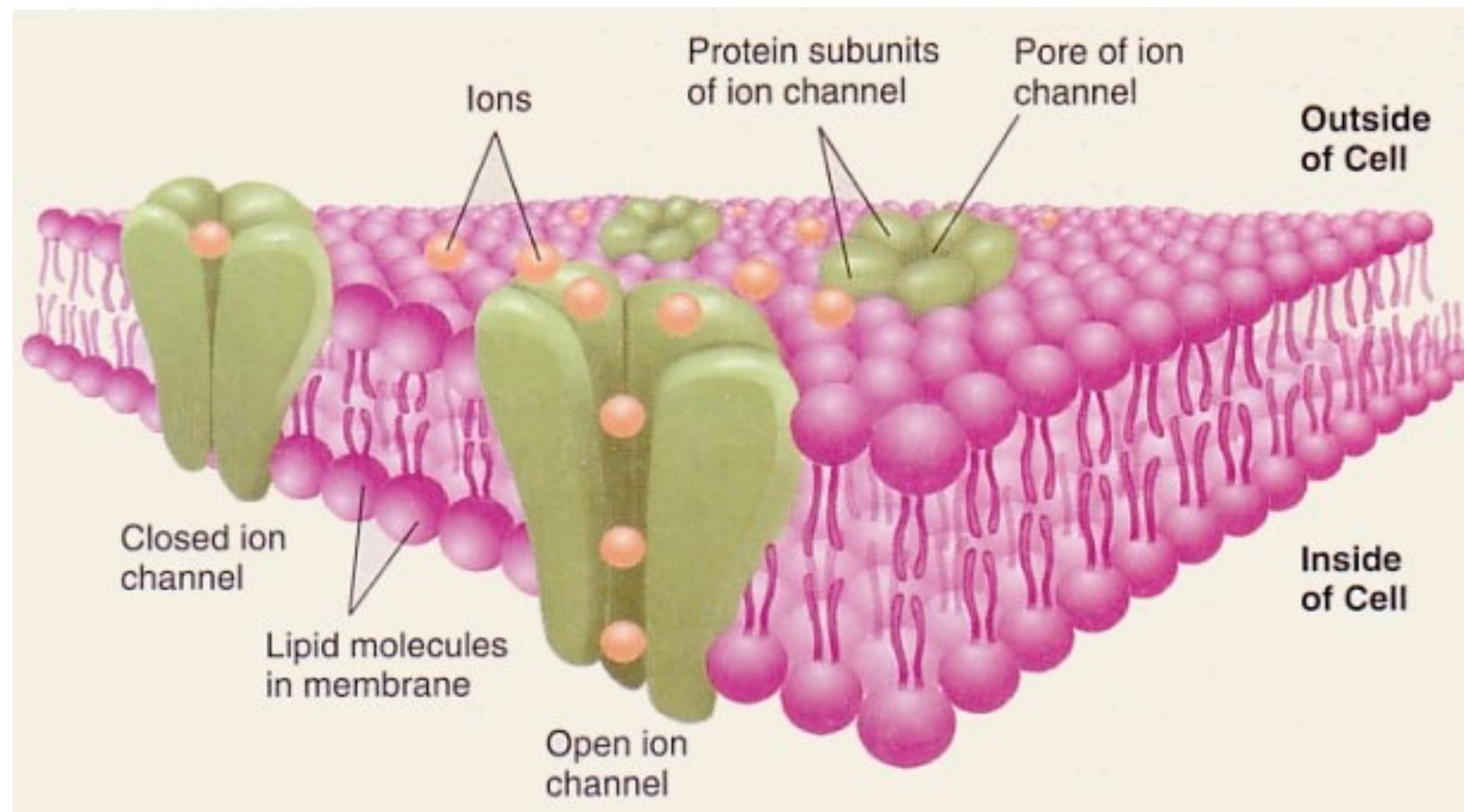


Single Neuron Computation



Course Curriculum

- Introduction to Computational Neuroscience
- Single Neuron Computation
- Connect Neurons to Perform Computation
- Recurrent network and Attractor Dynamics
- E/I balance and Chaotic Dynamics
- Efficient coding principle
- Stimulus discrimination, Fisher information and Information-limited correlation
- Normative approach to neural computation and the emergence of sensory representation
- Dimensionality reduction and motor representation
- How does an animal move, an integrated approach, and insights from *C. elegans*
- Reward and Reinforcement Learning

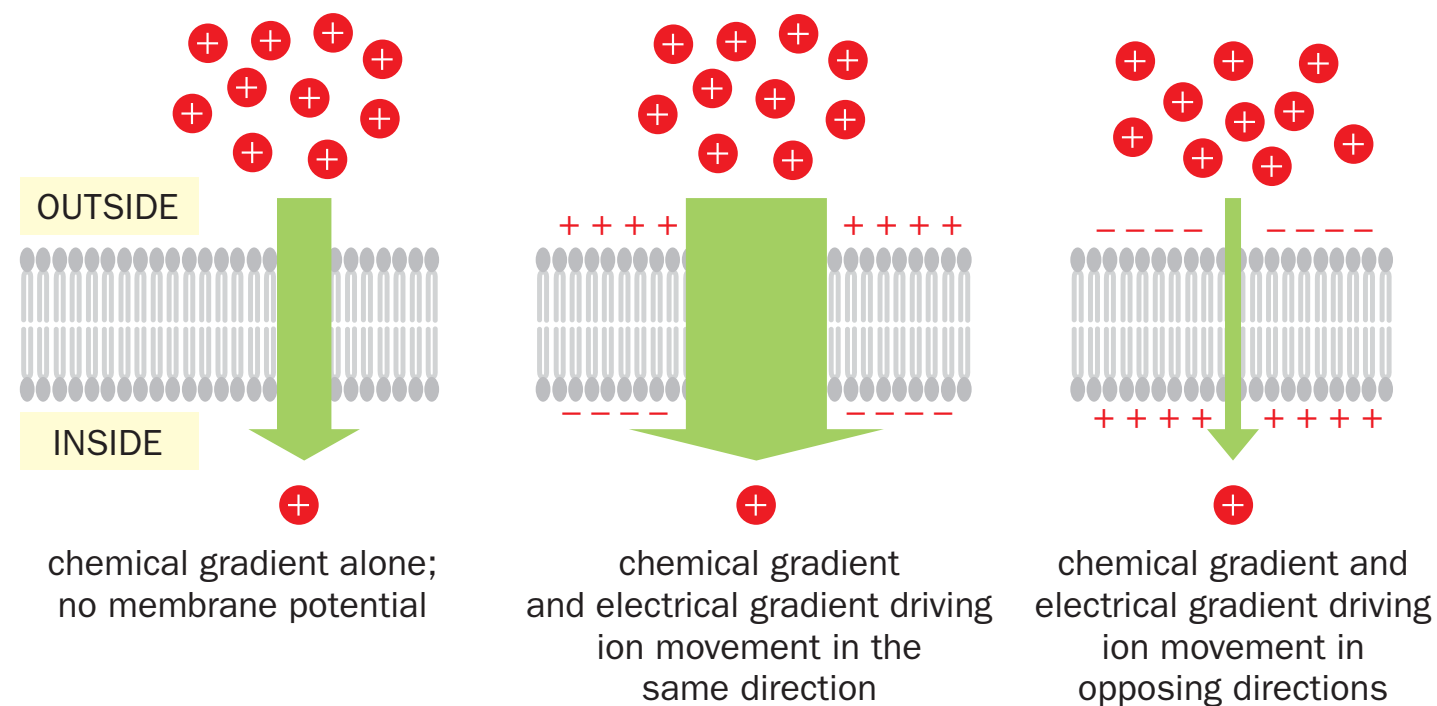


$$qV \sim k_B T$$

$$k_B = 8.6 \times 10^{-5} \text{ eV/K}$$

$$T = 300 \text{ K}$$

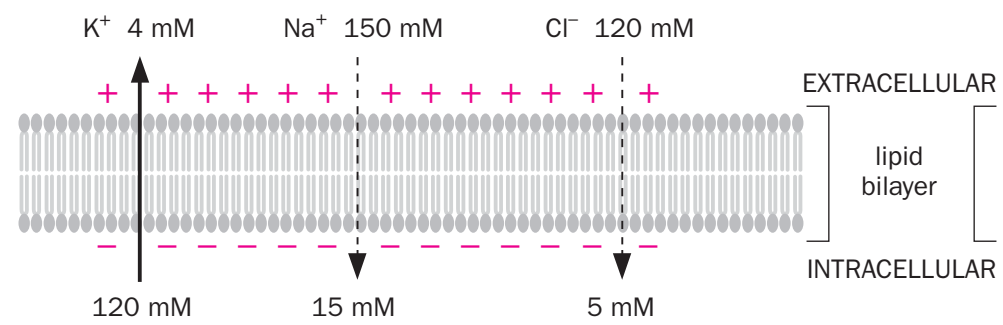
What determines the resting potential of a neuron?



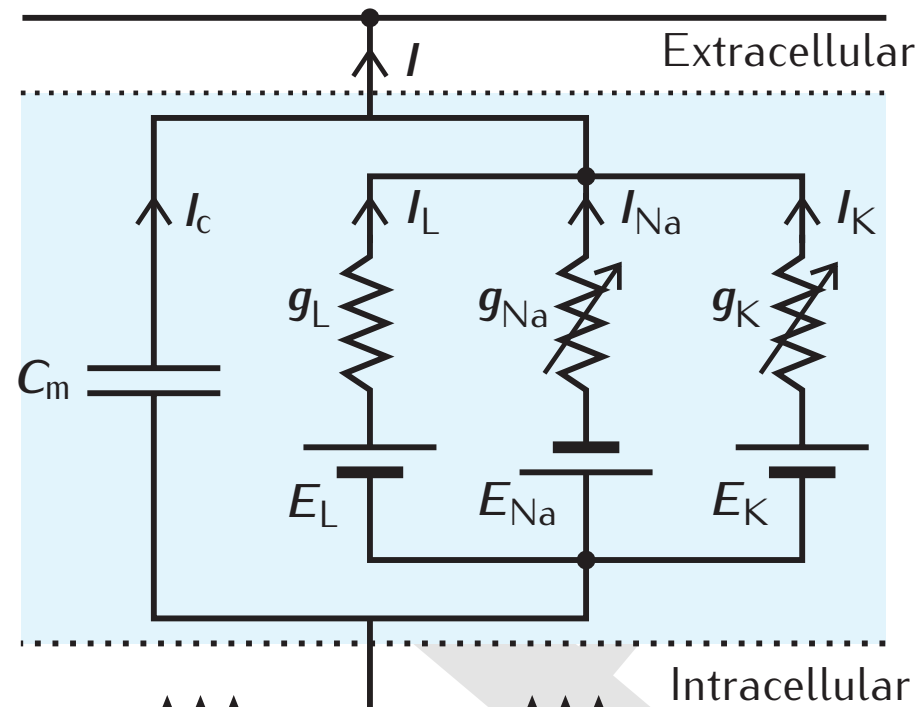
$$E = \frac{k_B T}{ze} \ln \frac{n_{out}}{n_{in}}$$

Ion pump

(A)



The Equivalent Electronic Circuit of a Neuron



$$C_m \frac{dV}{dt} = - \sum_i g_i(V)(V - E_i) - \bar{g}_L(V - E_L) - I_e$$

$$C_m \approx 0.1 - 1 \text{ nF}$$

$$r_m = 1/g \approx 10 - 100 \text{ M}\Omega$$

$$\tau_m = C_m r_m \approx 10 - 100 \text{ ms}$$

Squid Giant Axon

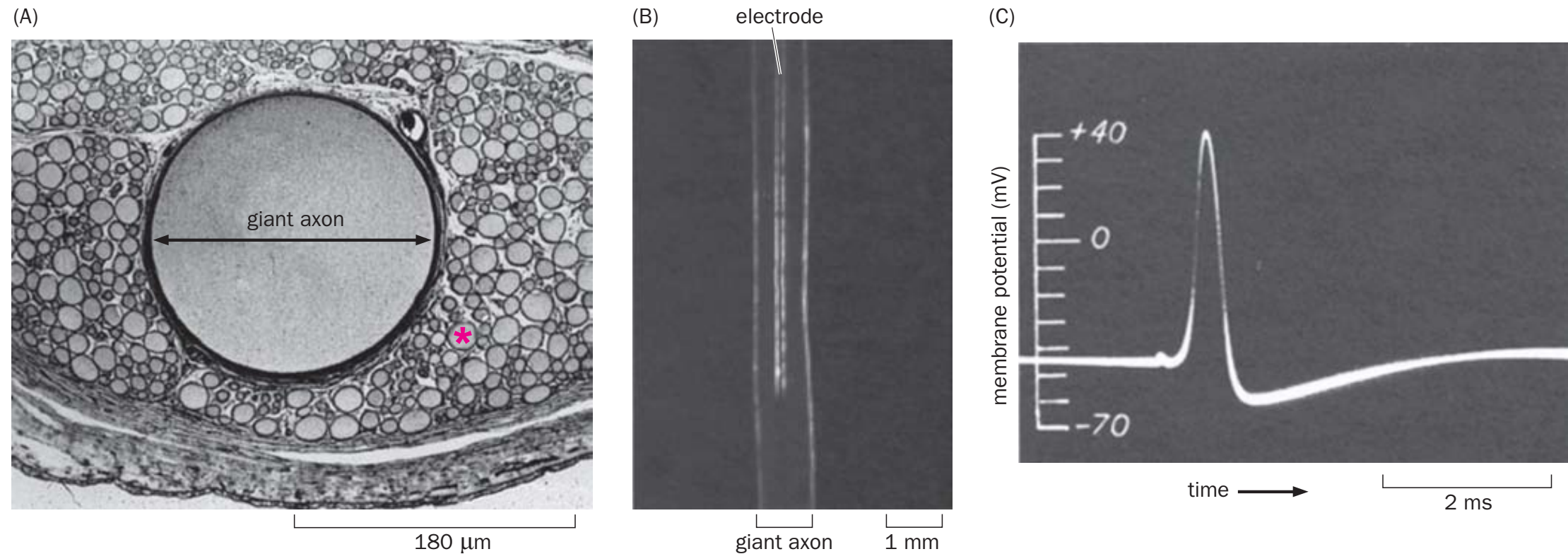
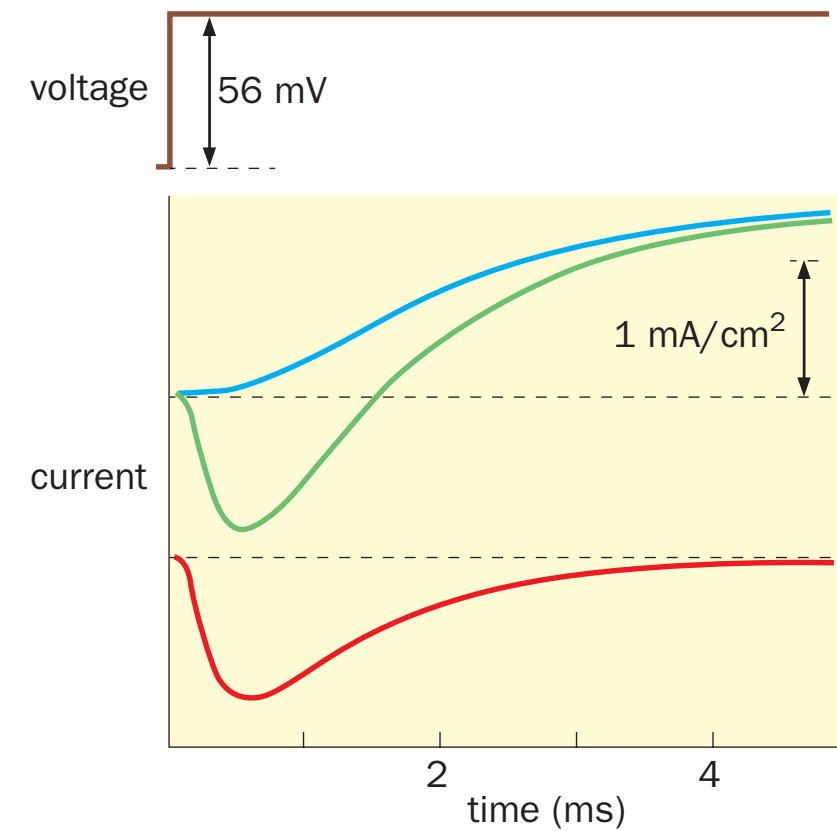
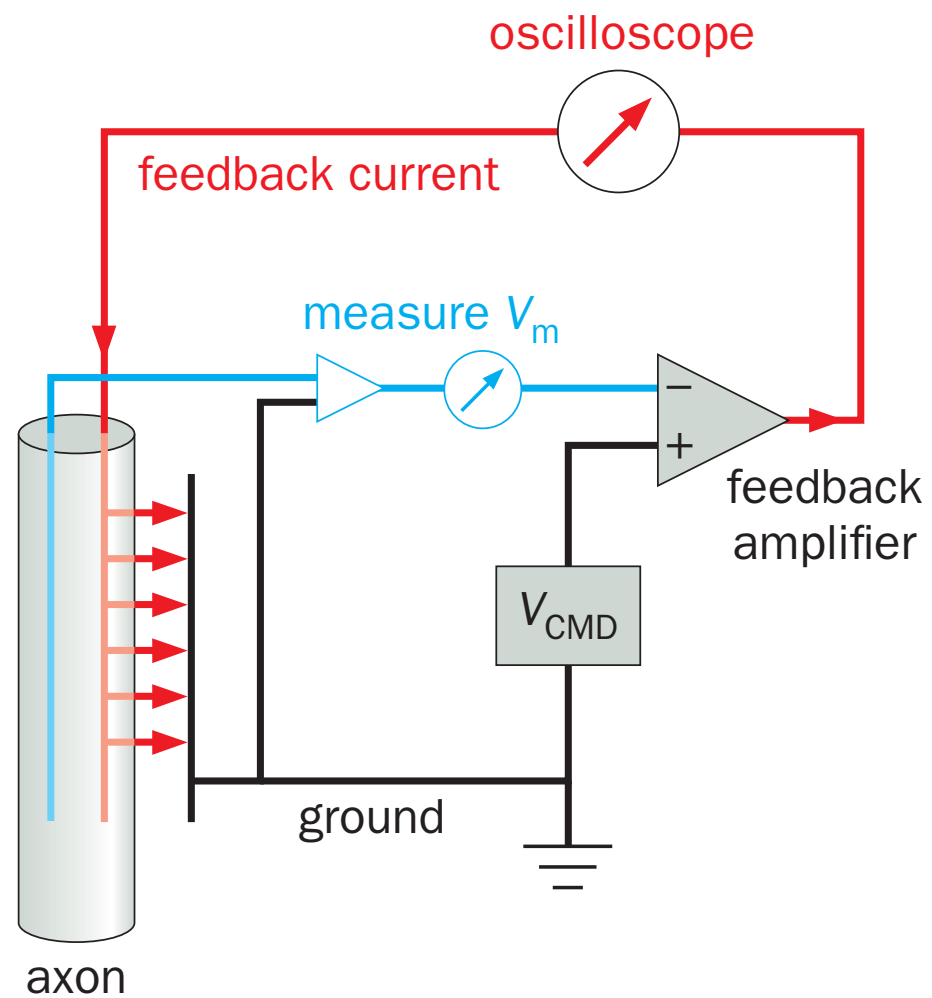
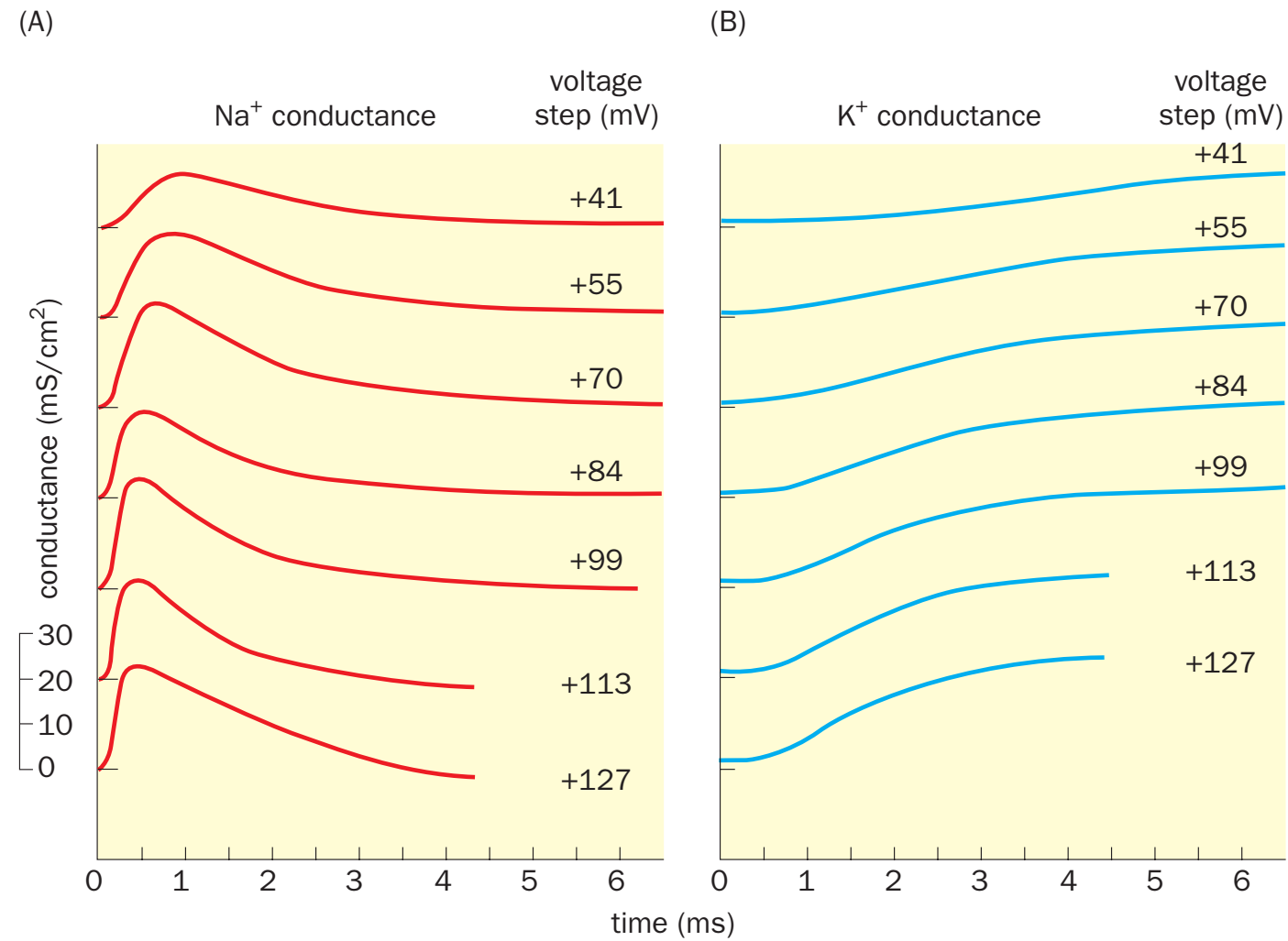


Figure 2-19 Studying action potentials in the squid giant axon. (A) Electron micrograph of a cross section of a squid giant axon showing its large diameter (~180 μm for this sample) as compared to neighboring axons (for example, the axon indicated by *). (B) Photograph of an electrode inserted inside a squid giant axon whose diameter is close to 1 mm. (C) An action potential recorded from the squid giant axon. (A, courtesy of Kay Cooper and Roger Hanlon; B, from Hodgkin AL & Keyes RD [1956] *J Physiol* 131:592–616; C, from Hodgkin AL & Huxley AF [1939] *Nature* 144:710–711. With permission from Macmillan Publishers Ltd.)

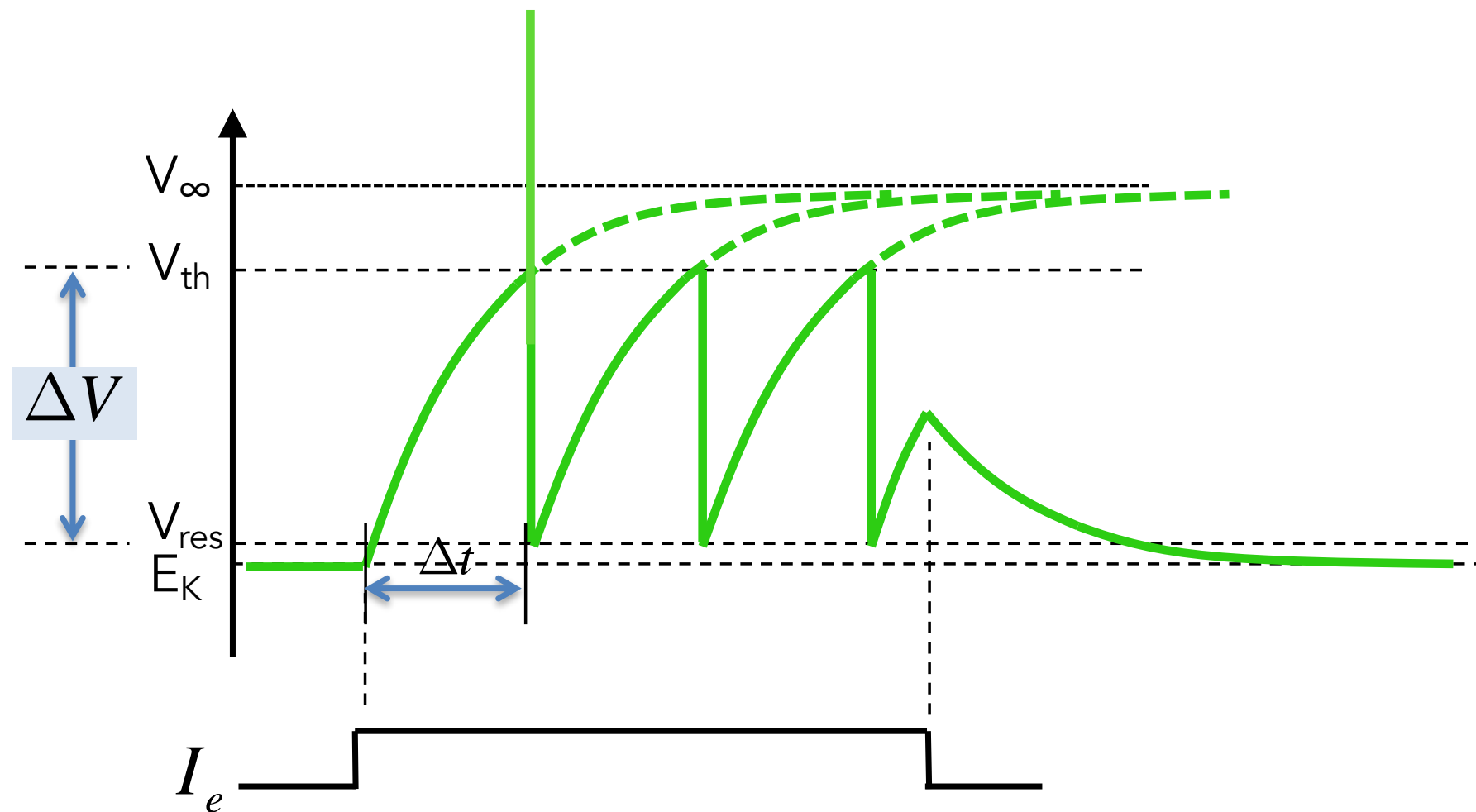
Voltage Clamp Recording



Voltage-gated Conductance



Integrate-and-Fire model

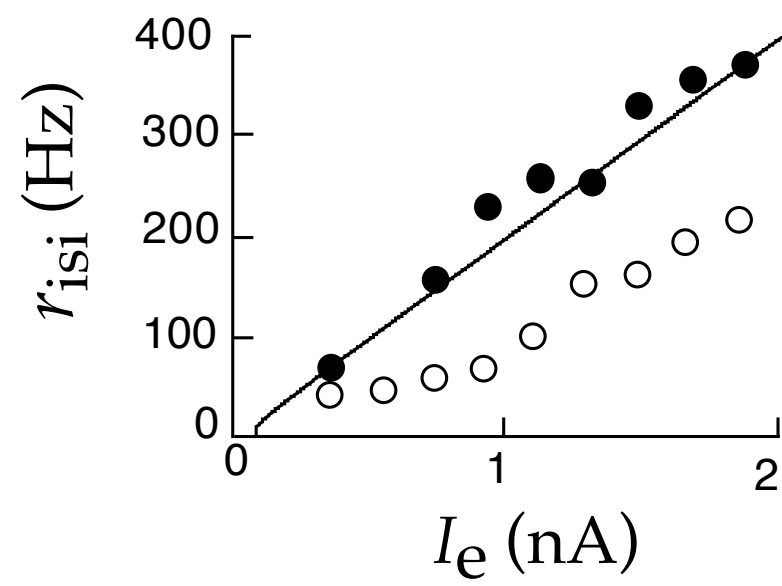


$$C \frac{dV}{dt} = -g(V - E_K) + I_e$$

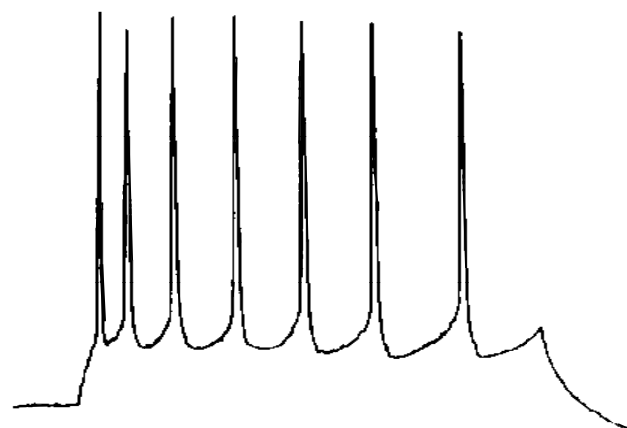
$$V(t_{spike}^-) = V_{th}$$

$$V(t_{spike}^+) = V_{res}$$

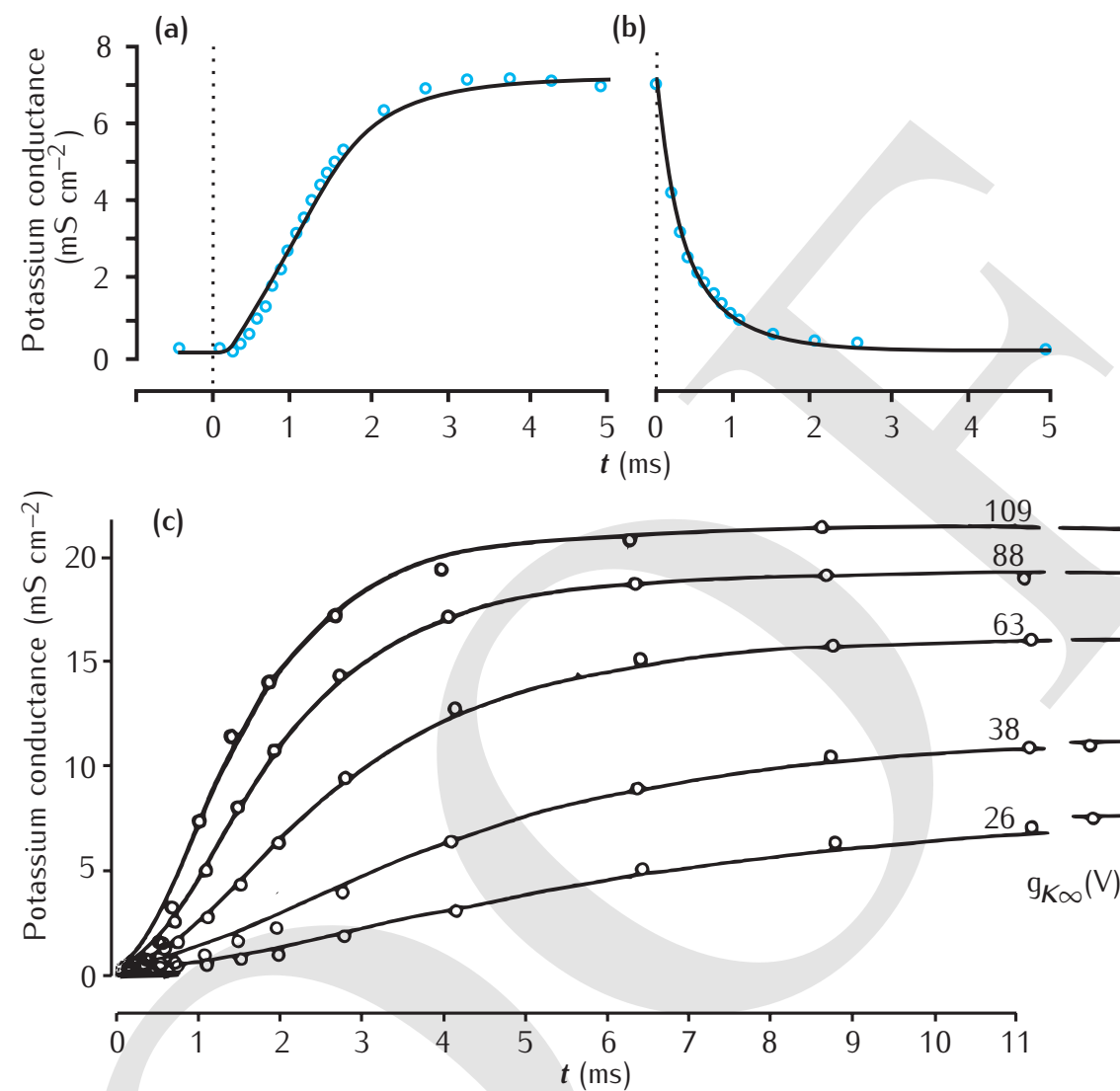
A



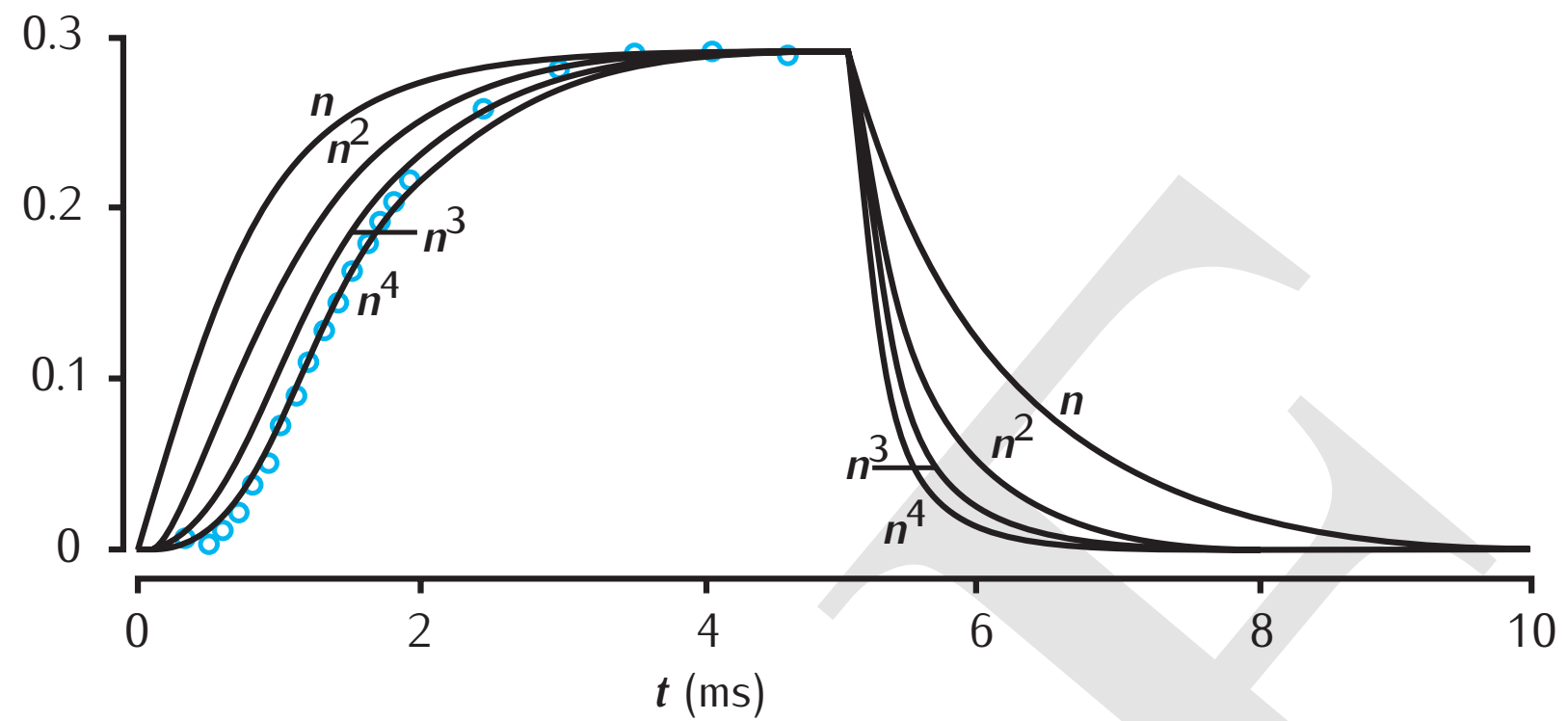
B

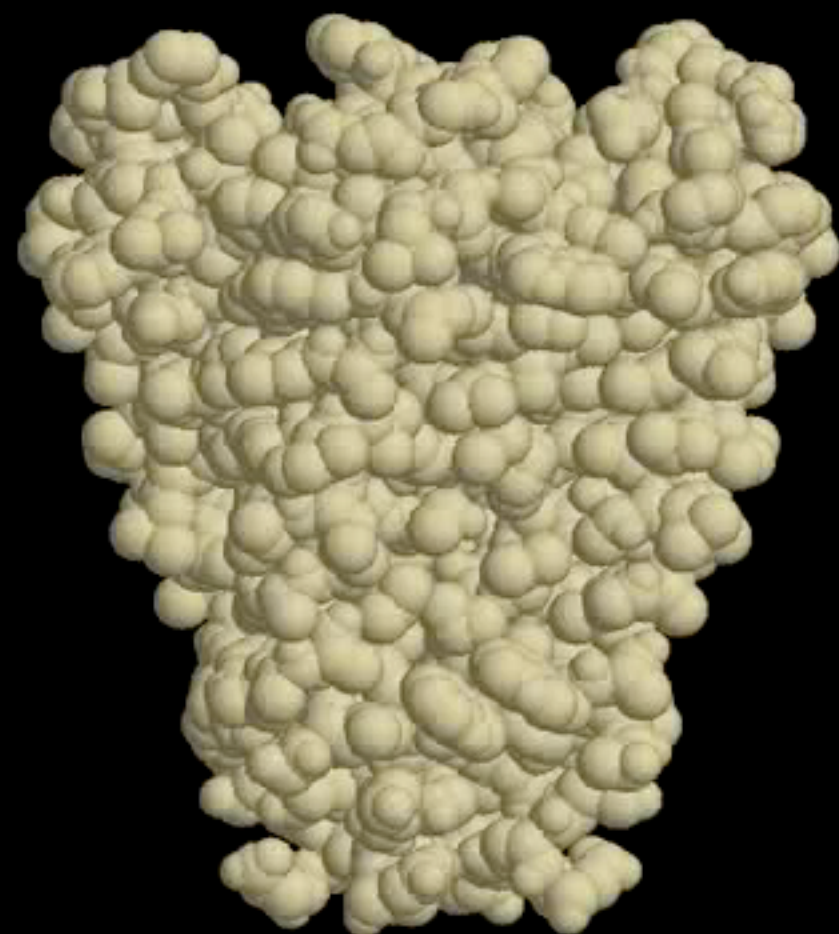


Voltage-gated Conductance of K^+

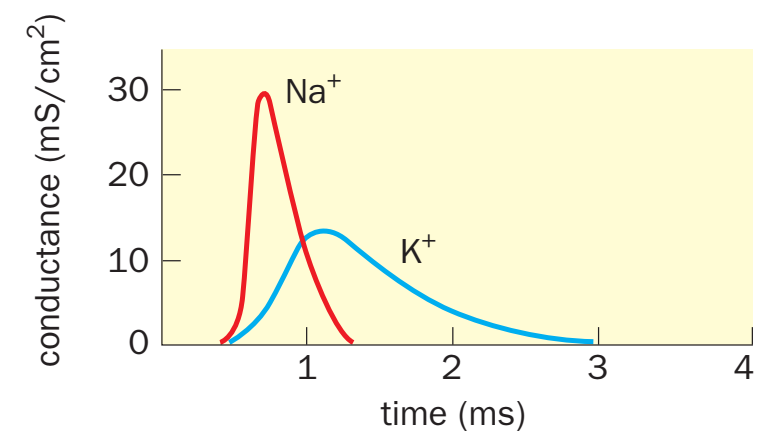
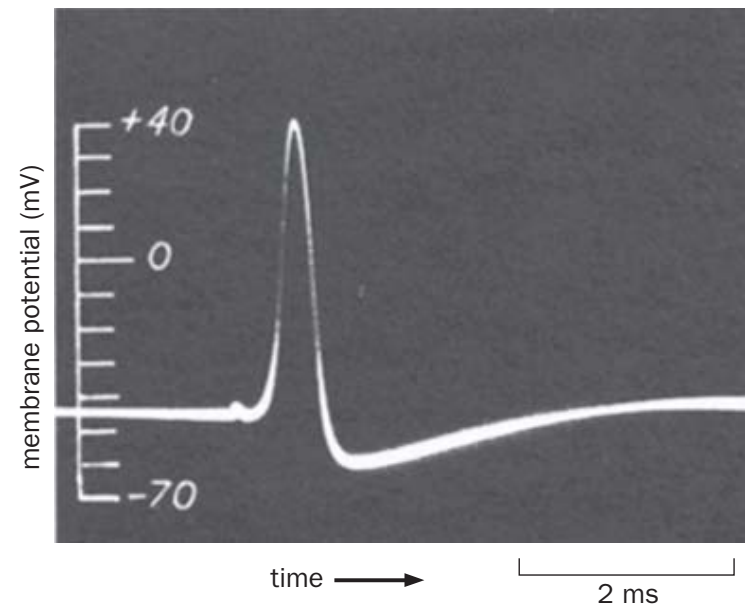


Voltage-gated Conductance of K^+



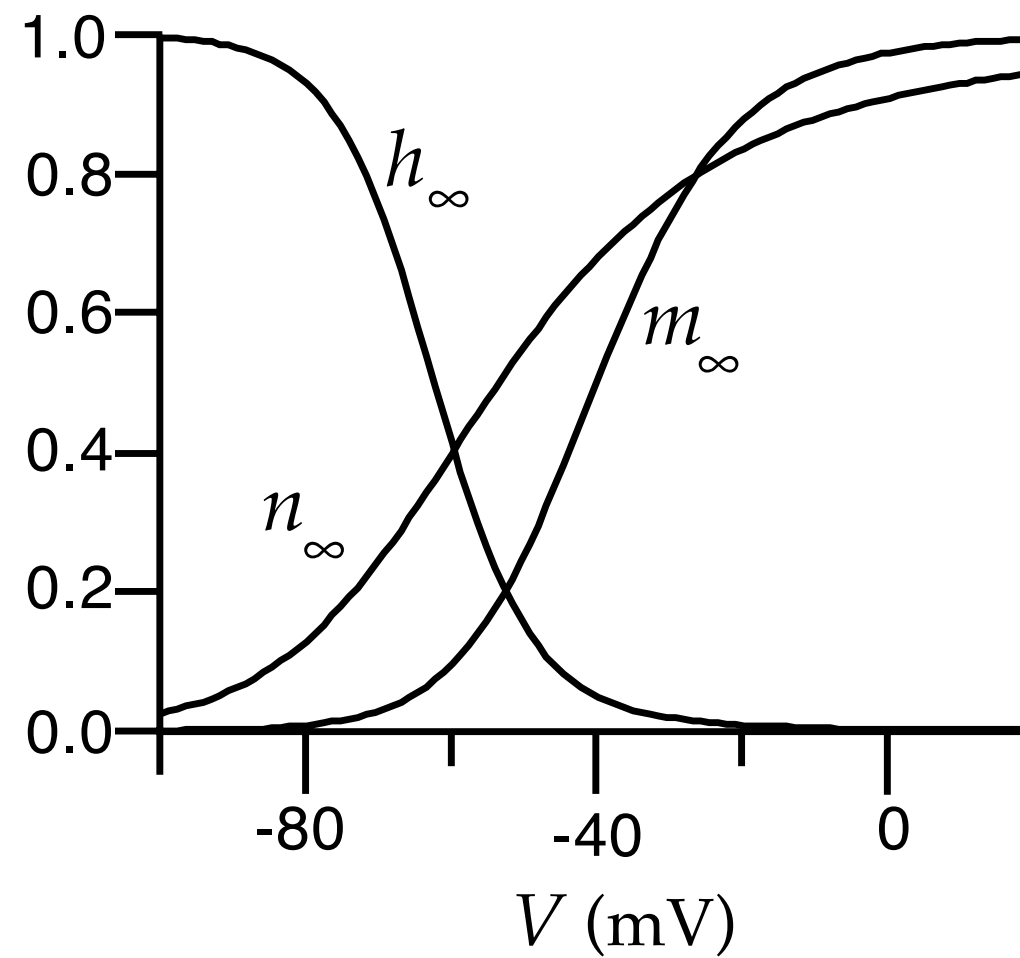


(c)



$$C \frac{dV}{dt} = -g_K n^4 (V - E_K) - g_{Na} m^3 h (V - E_{Na}) - g_L (V - E_L) - I_e$$

Transient Na⁺ channel conductance

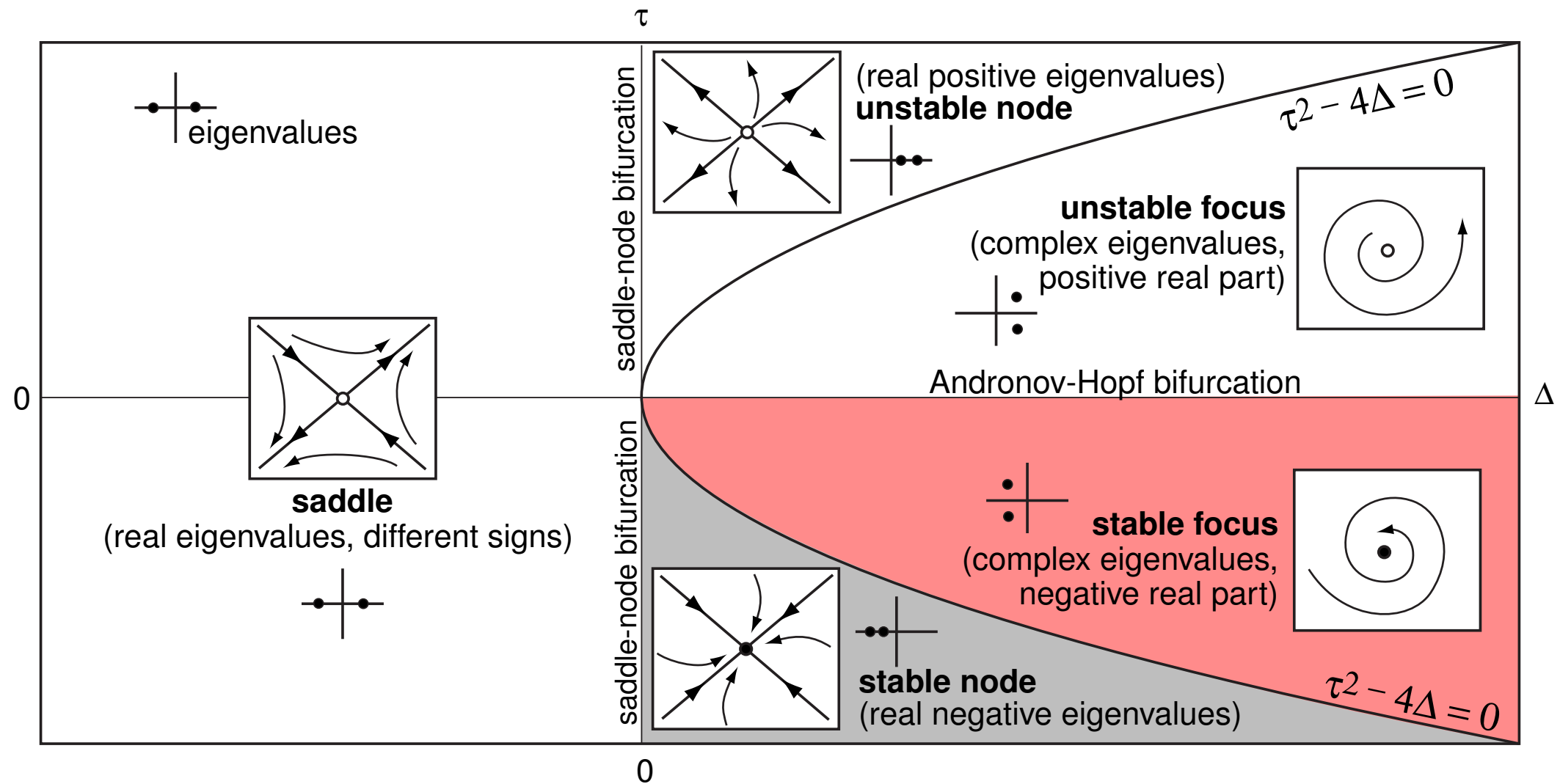


Fix Point one dimension

$$\frac{dV}{dt} = F(V)$$

$$F(V^*) = 0$$

Fix Point in two dimensions

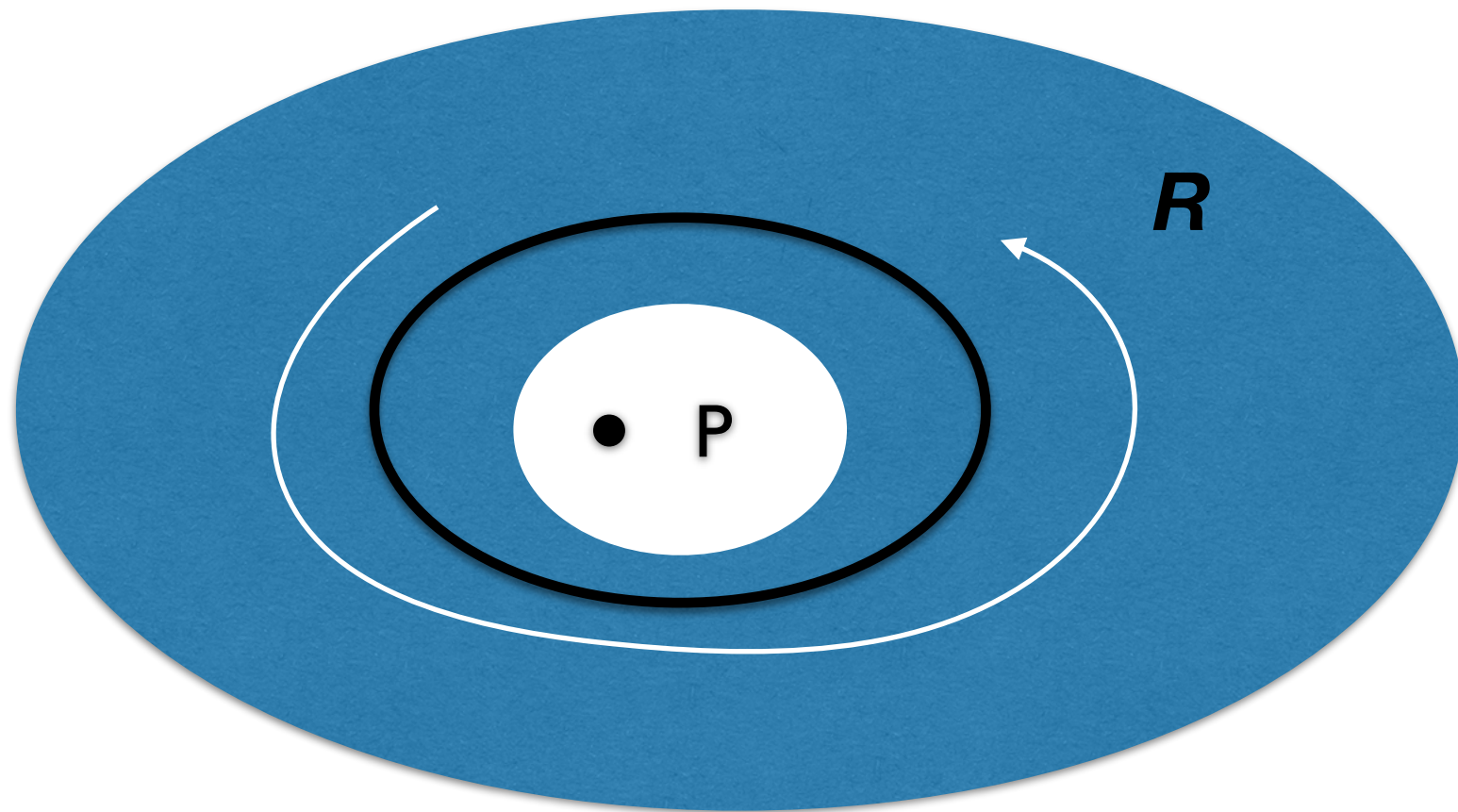


Poincare-Bendixson Theorem (2D)

- R is a closed, bounded subset of the plane;
- $\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x})$ is a continuously differentiable vector field on an open set containing R ;
- R does not contain any fixed points; and
- There exists a trajectory C that is “confined” in R , in the sense that it starts in R and stays in R for all future time

Then either C is a closed orbit, or it spirals toward a closed orbit at $t \rightarrow \infty$. In either case, R contains a closed orbit.

Poincare-Bendixson Theorem (2D)



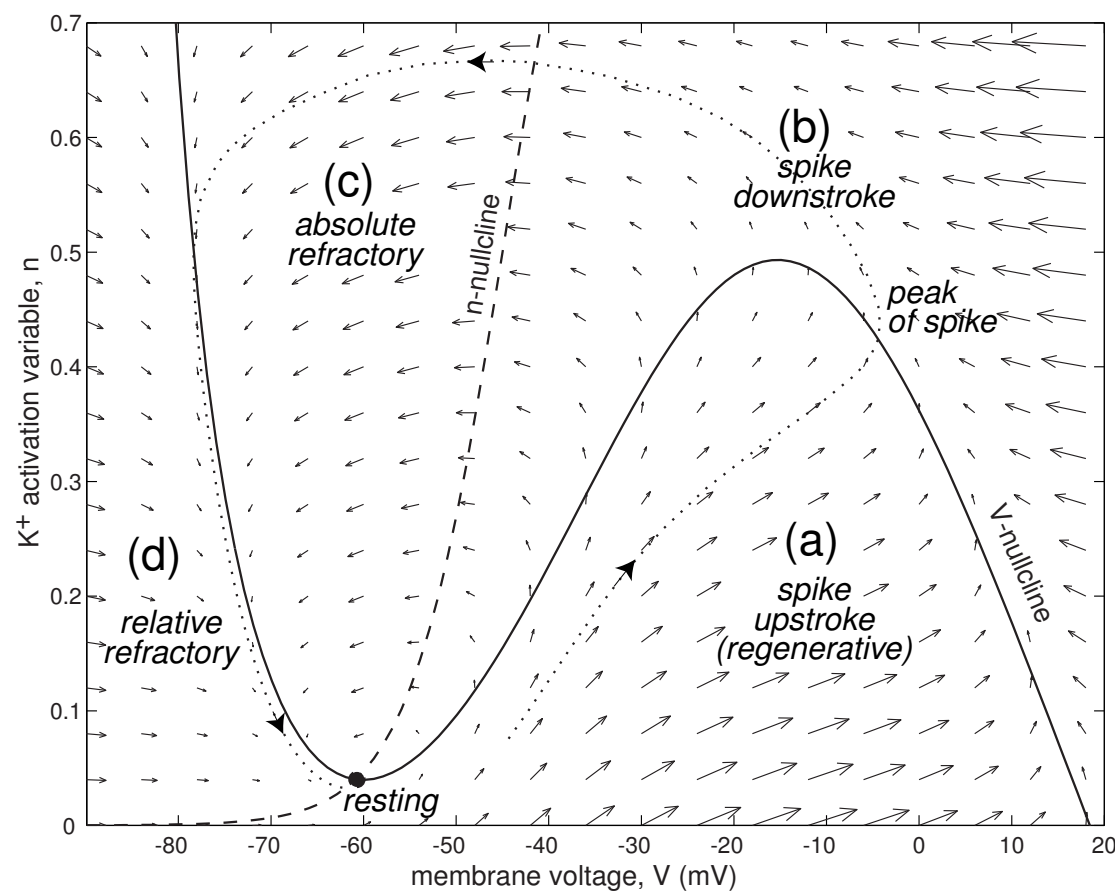


Figure 4.4: Nullclines of the $I_{Na,p}+I_K$ -model (4.1, 4.2) with low-threshold K^+ current in Fig.4.1b. (The vector field is slightly distorted for the sake of clarity of illustration).

Dynamic systems in Neuroscience

Eugene Izhikevich 2007

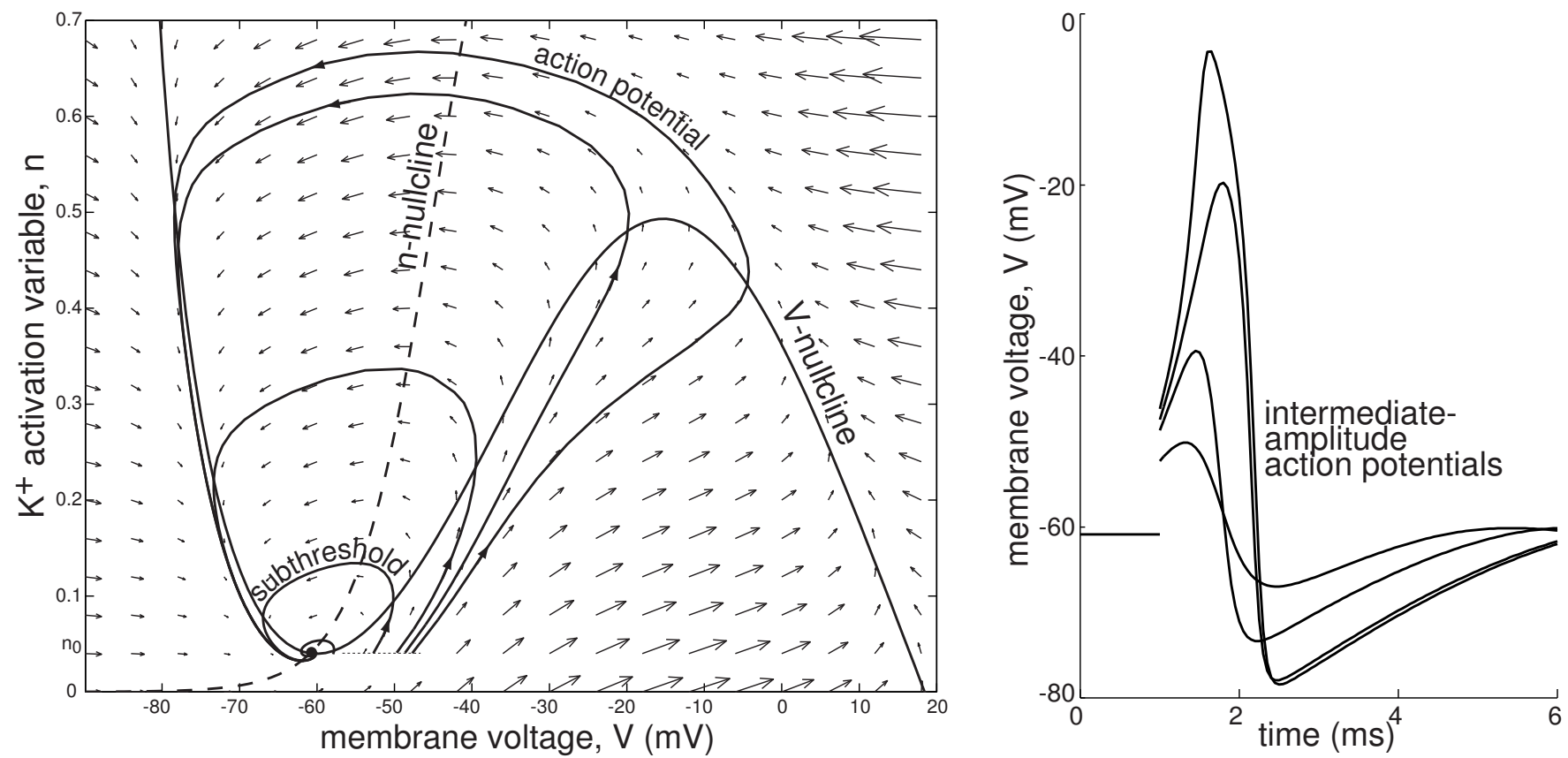
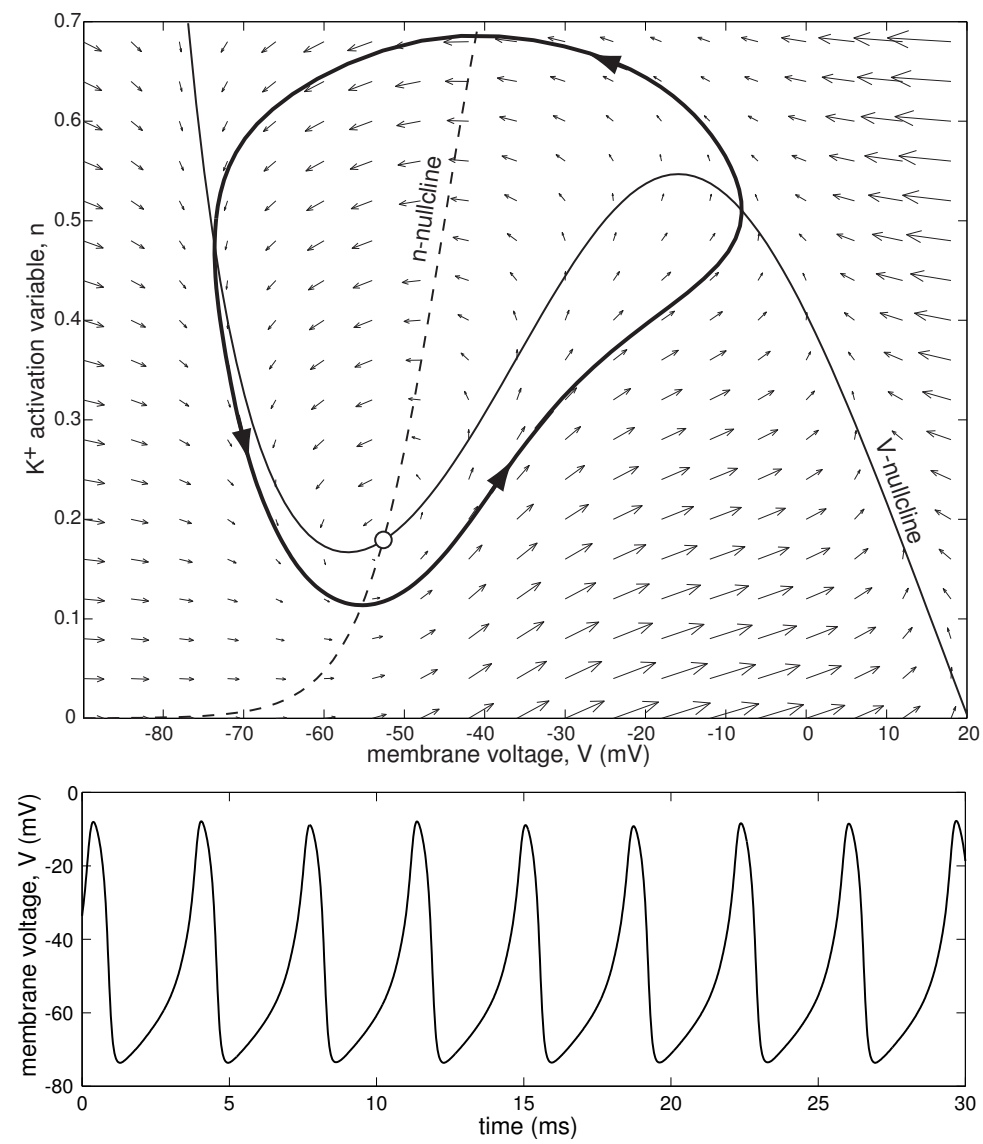


Figure 4.7: Failure to generate all-or-none action potentials in the $I_{Na,p} + I_K$ -model.

Dynamic systems in Neuroscience

Eugene Izhikevich 2007

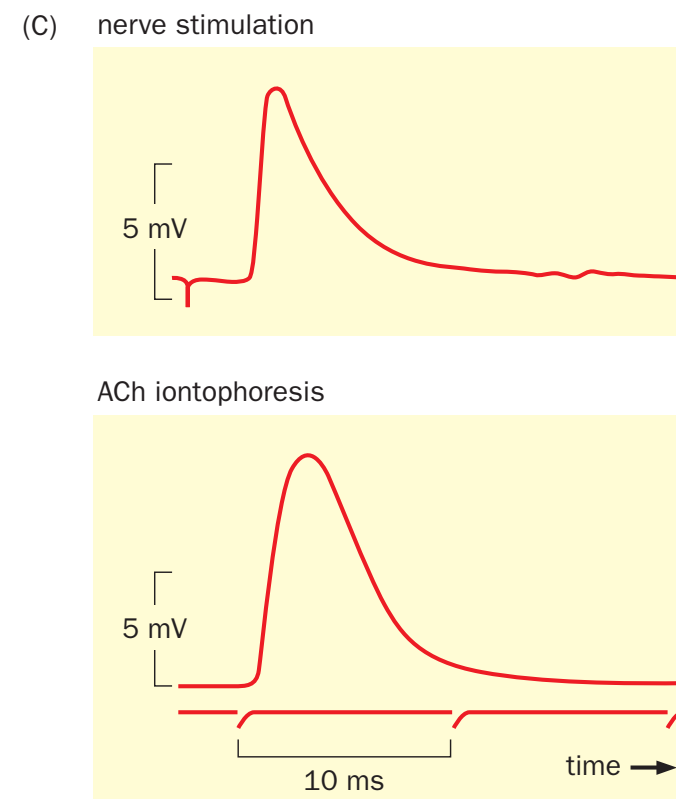
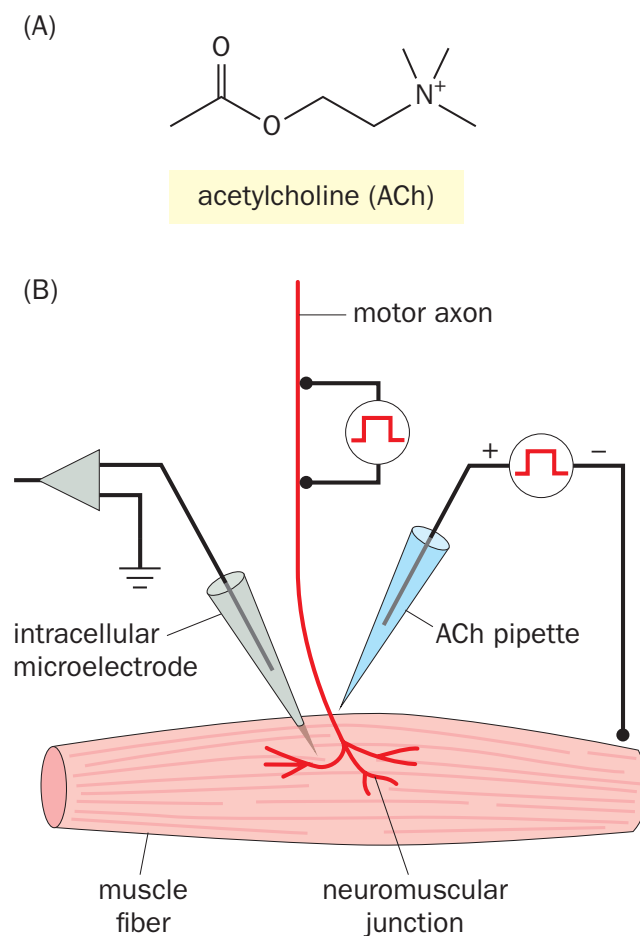


Dynamic systems in Neuroscience

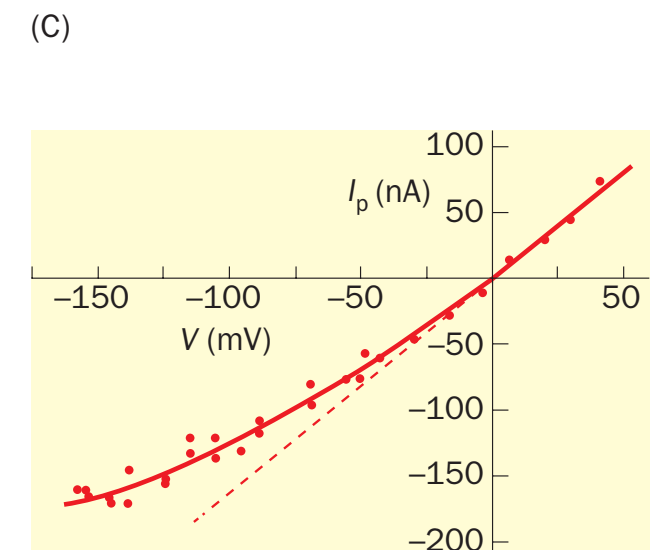
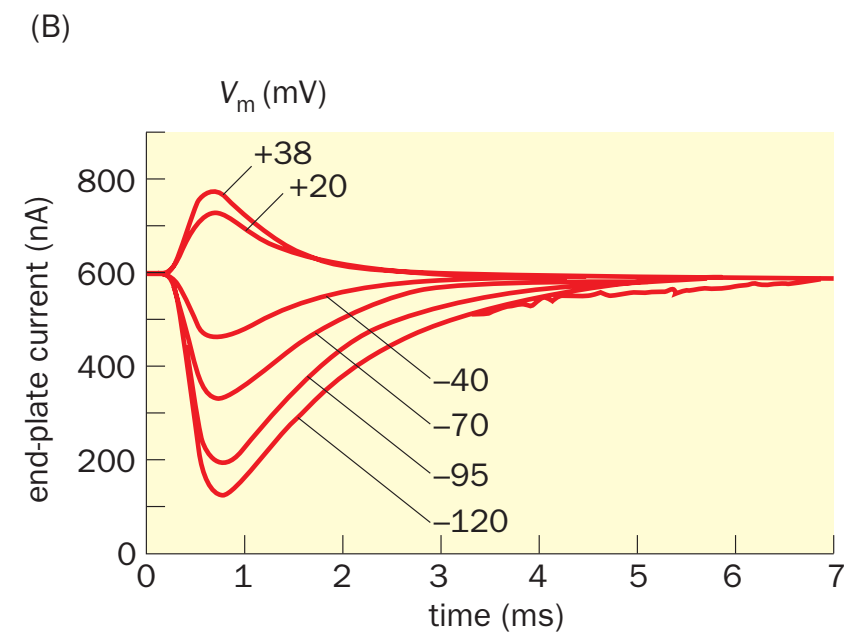
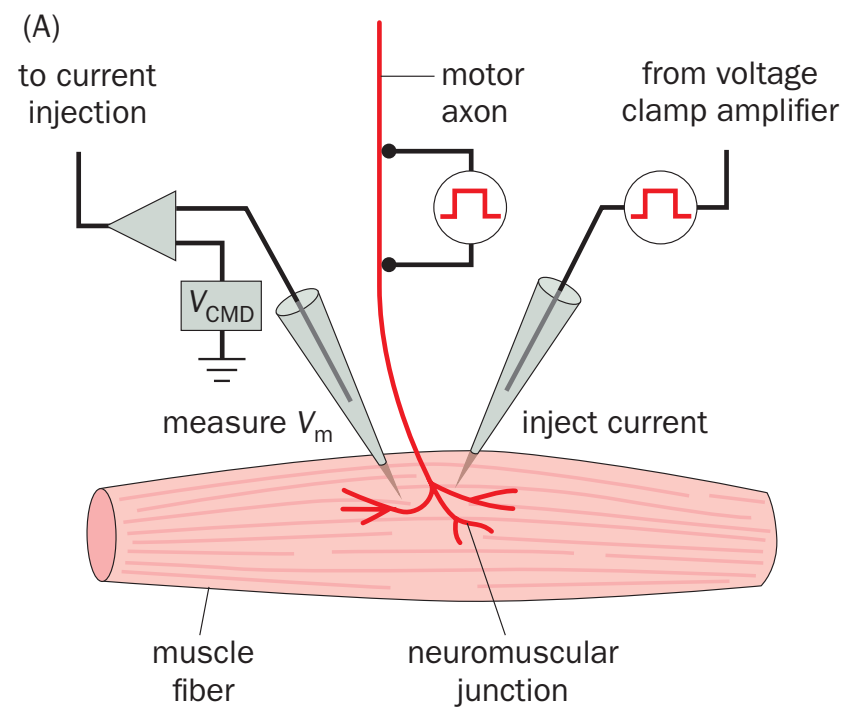
Eugene Izhikevich 2007

Synaptic Transmission

Neurotransmitter release evokes membrane potential change in the postsynaptic neuron



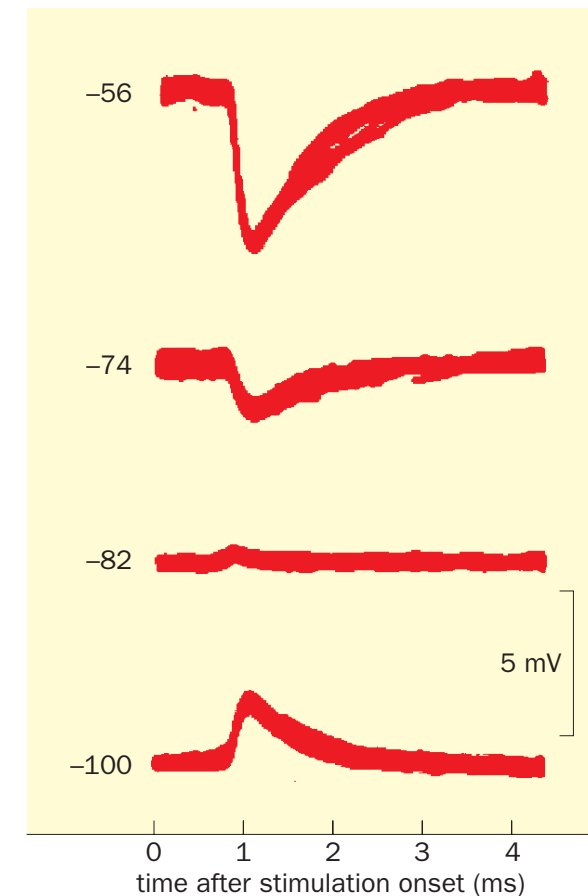
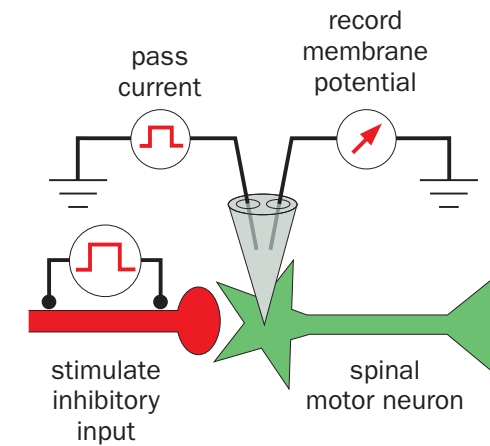
Reversal potential of a synapse



Excitatory and Inhibitory Synapses

Table 3–2: Commonly used neurotransmitters

Neurotransmitter	Major uses in the vertebrate nervous system ¹
Acetylcholine	motor neurons that excite muscle; ANS ² neurons; CNS excitatory and modulatory neurons
Glutamate	most CNS excitatory neurons; most sensory neurons
GABA	most CNS inhibitory neurons
Glycine	some CNS inhibitory neurons (mostly in the brainstem and spinal cord)
Serotonin (5-HT)	CNS modulatory neurons
Dopamine	CNS modulatory neurons
Norepinephrine	CNS modulatory neurons; ANS ² neurons
Histamine	CNS modulatory neurons
Neuropeptides	usually co-released from excitatory, inhibitory, or modulatory neurons; neurosecretory cells

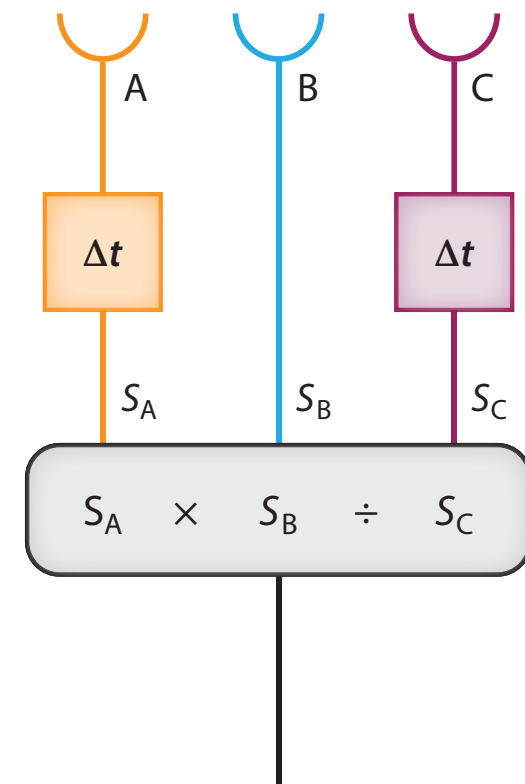
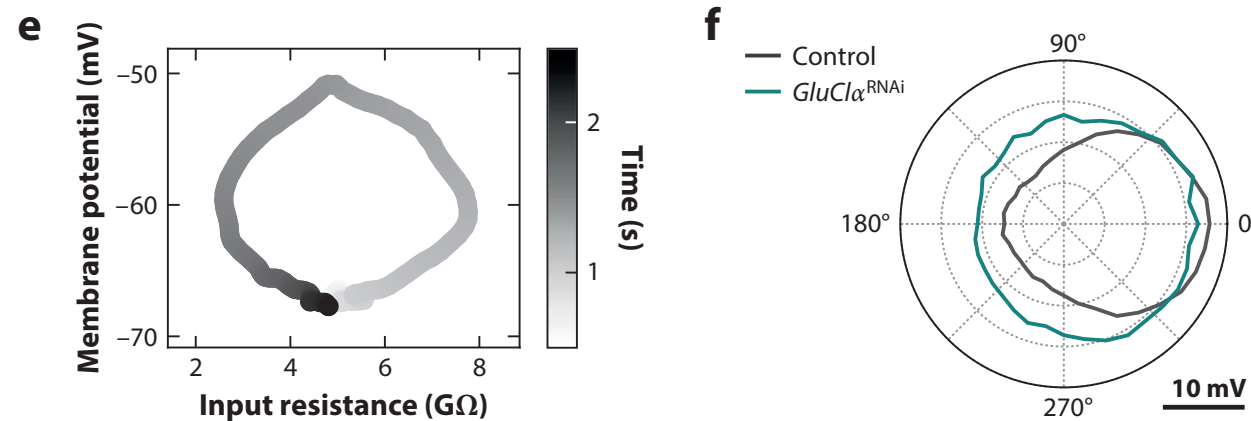
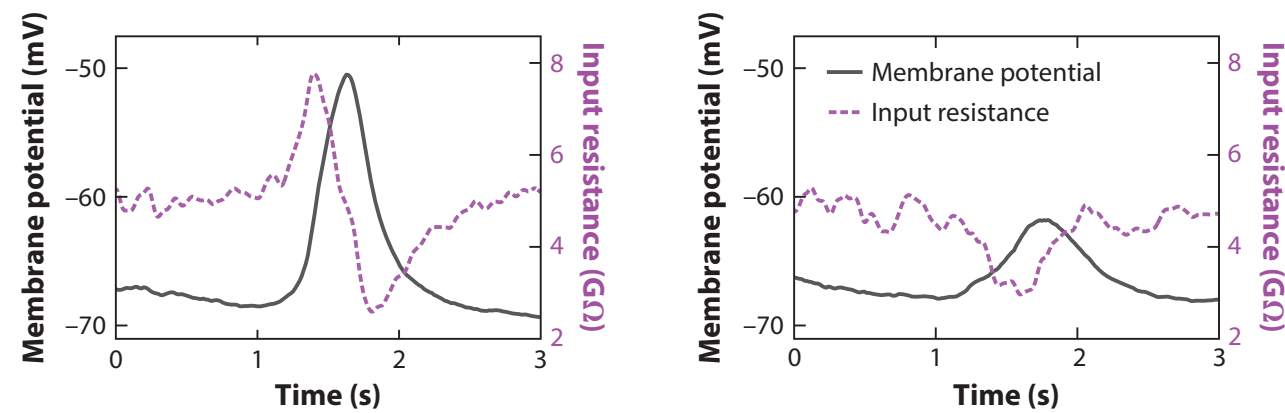
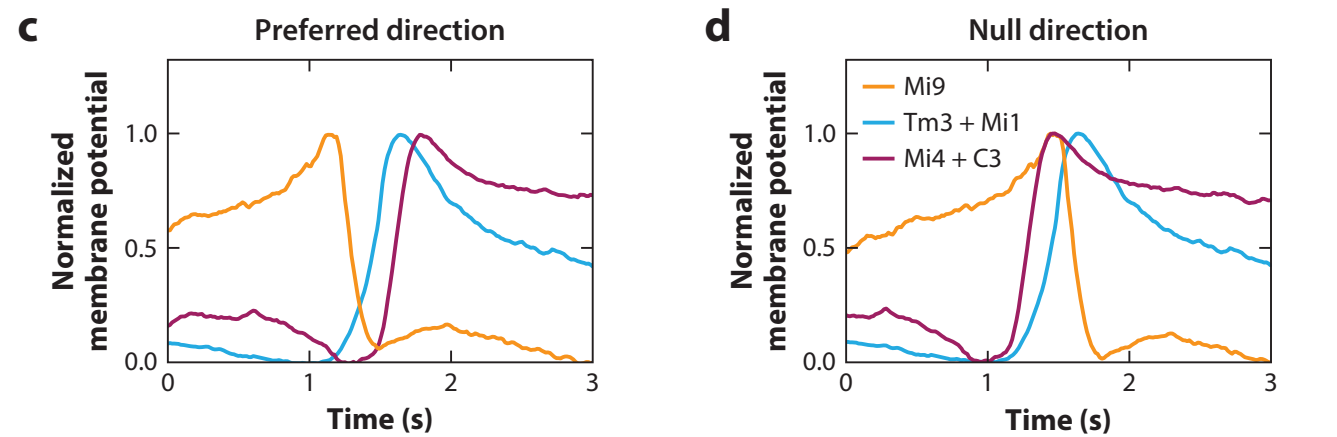
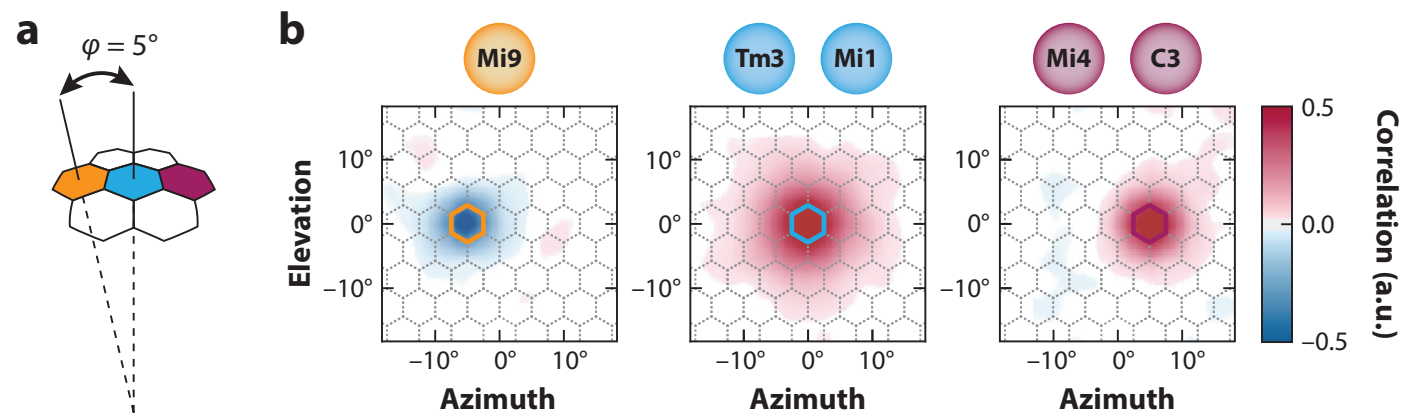


Shunting inhibition and divisive operation

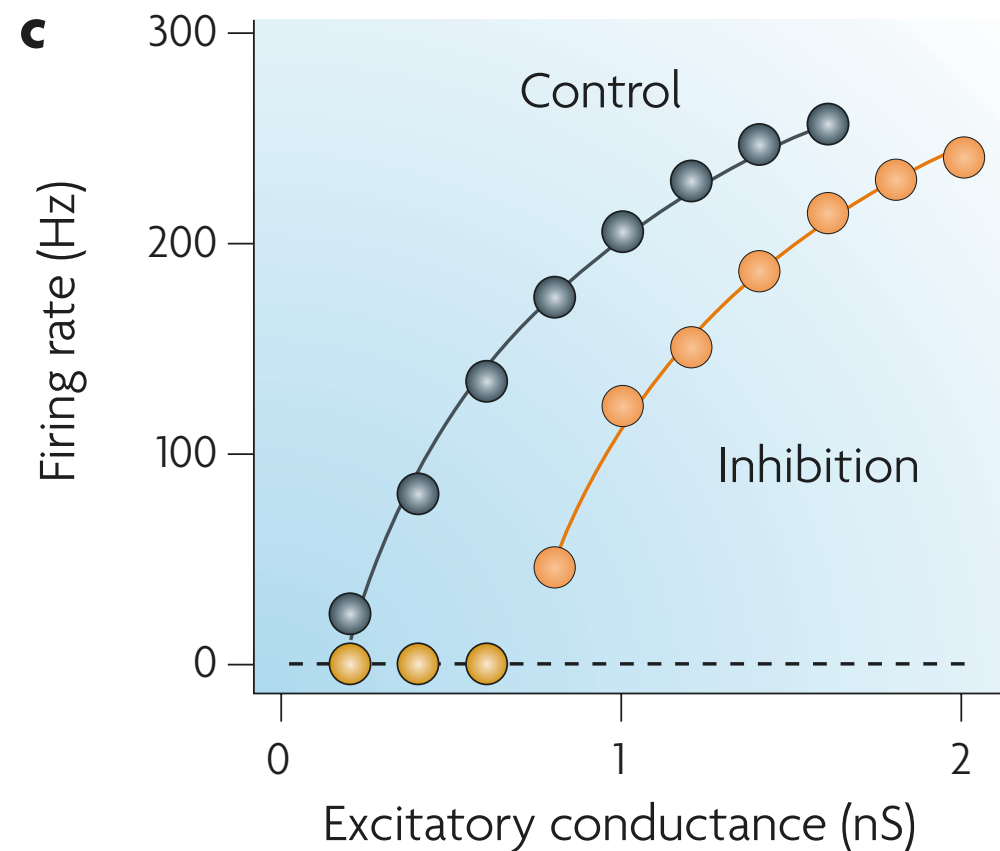
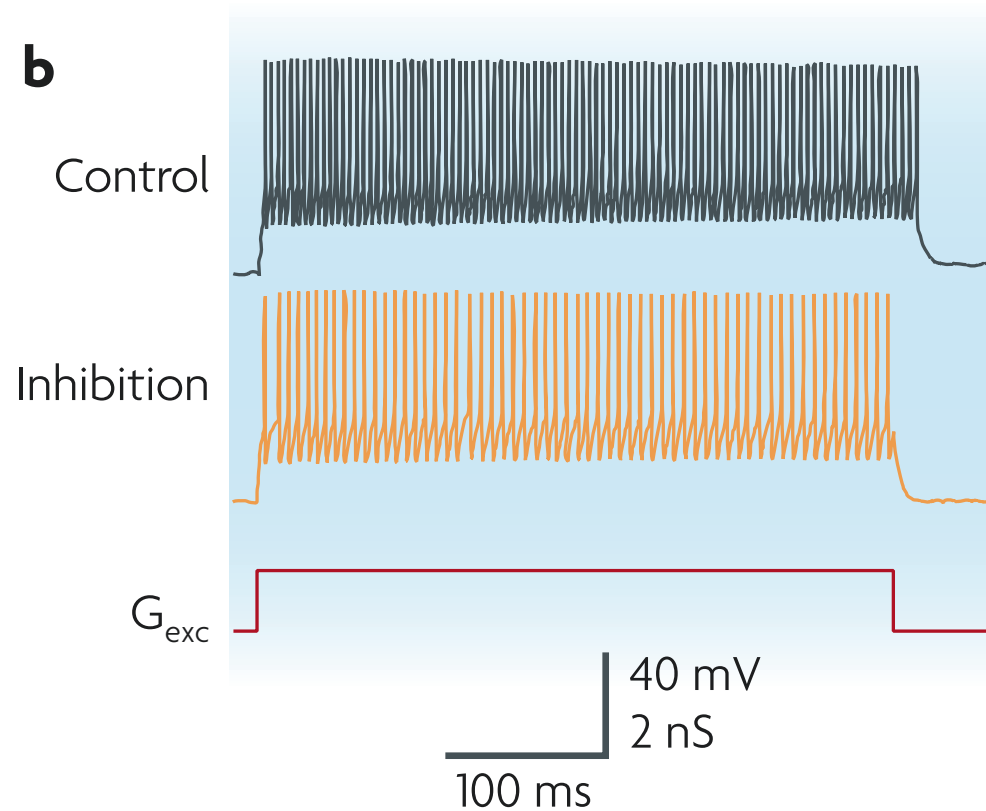
$$C_m \frac{dV}{dt} = -g_L(V - E_L) - g_{inh}(V - E_L) - g_{exc}(V - E_0)$$

$$V = \frac{(g_L + g_{inh})E_L + g_{exc}E_0}{g_L + g_{inh} + g_{exc}}$$

$$g_{inh} \gg g_L + g_{exc} \quad V \approx E_L + \frac{g_{exc}}{g_{inh}}E_0$$

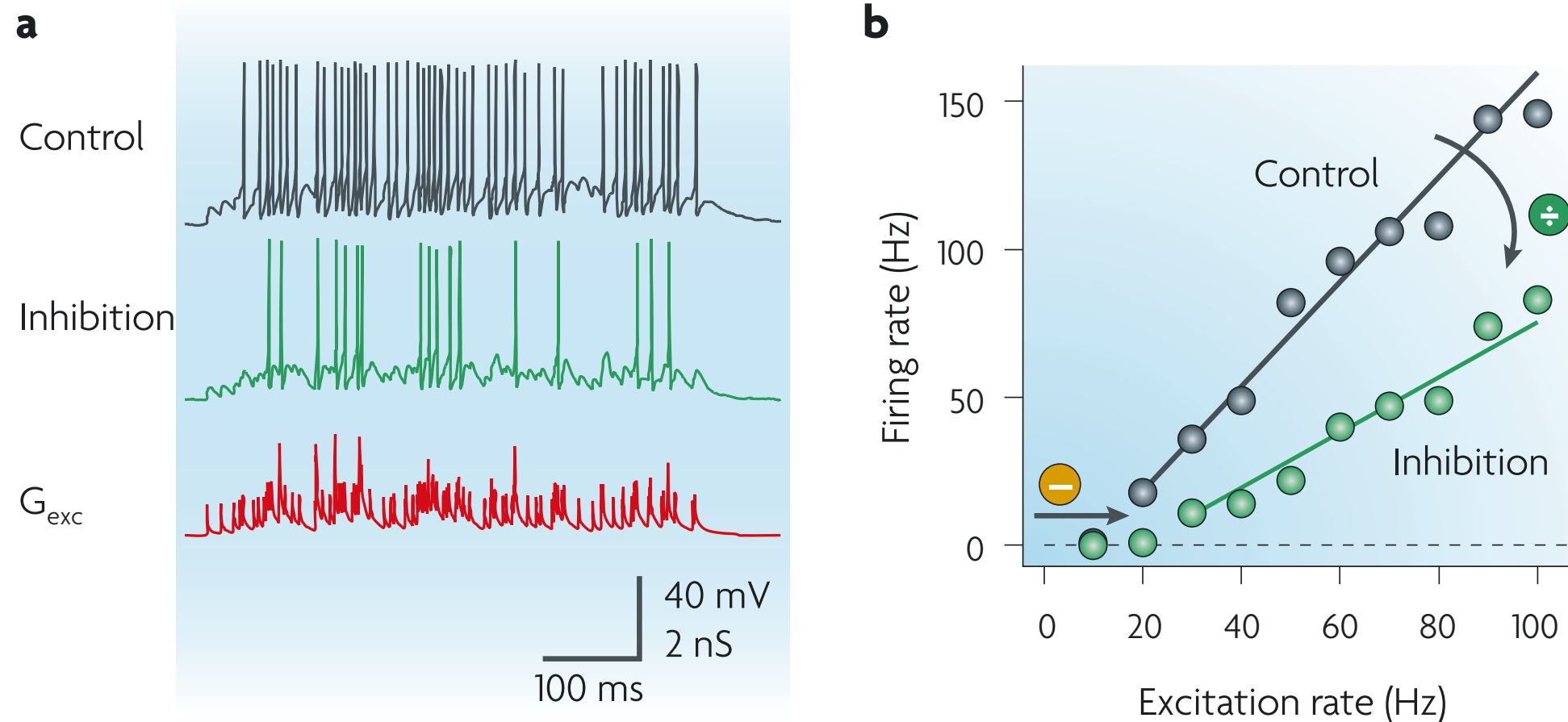


However, shunting inhibition becomes subtractive for firing rate

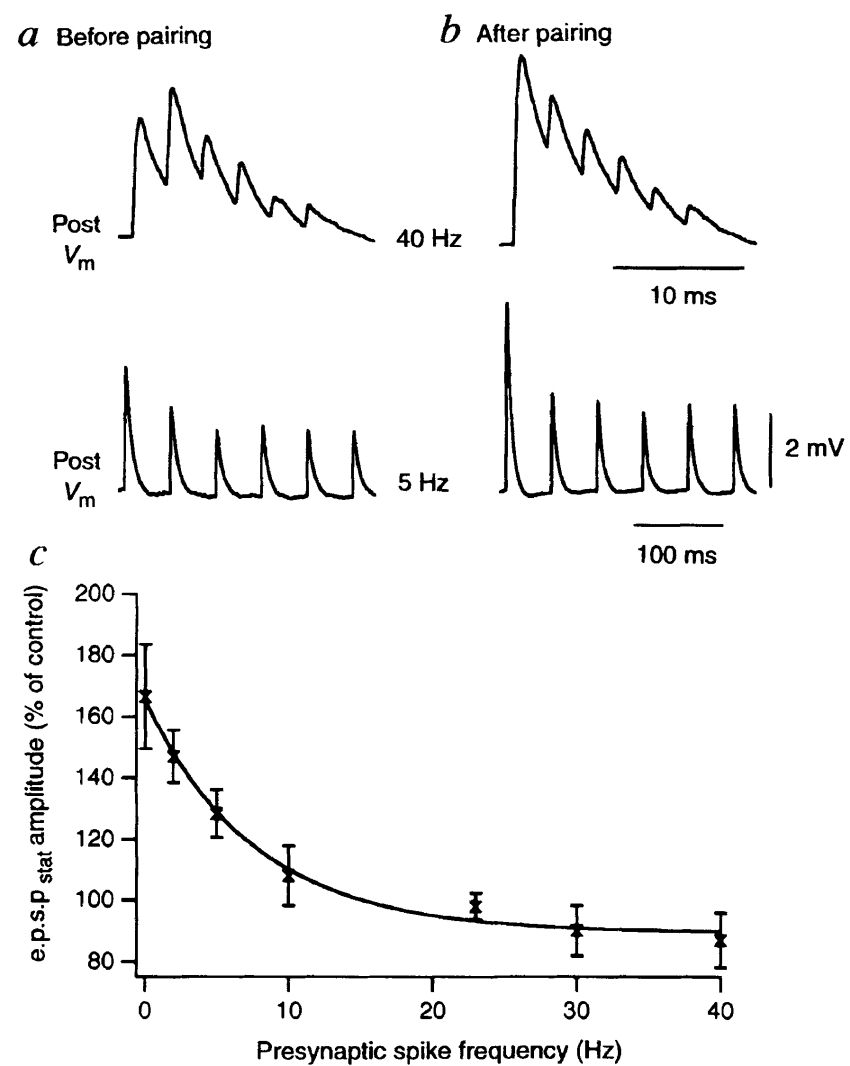
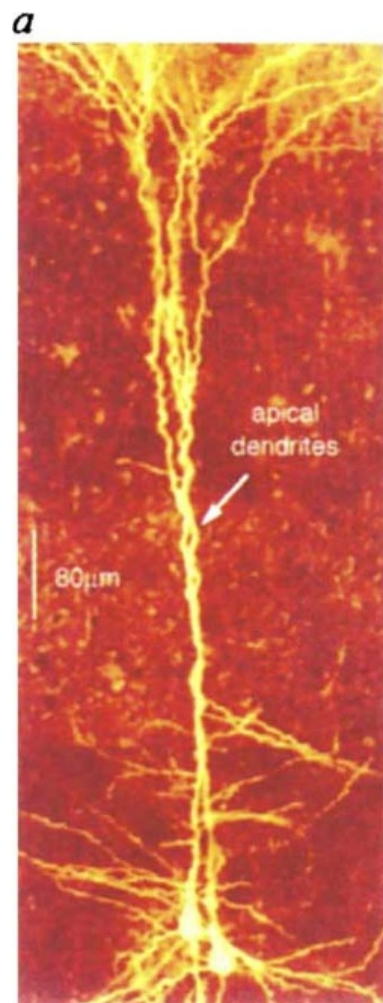


Why? (Your homework)

But Shunting inhibition can become divisive in the presence of large synaptic input noise

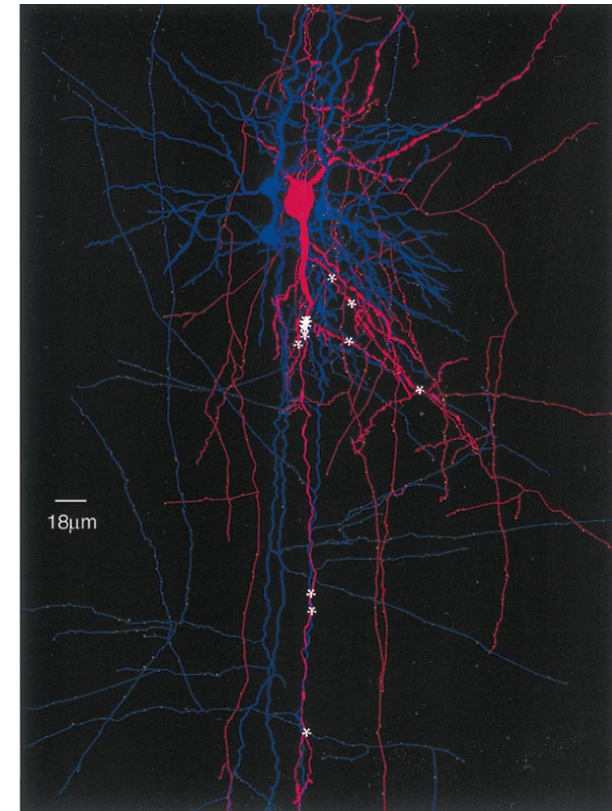


Short-term synaptic plasticity

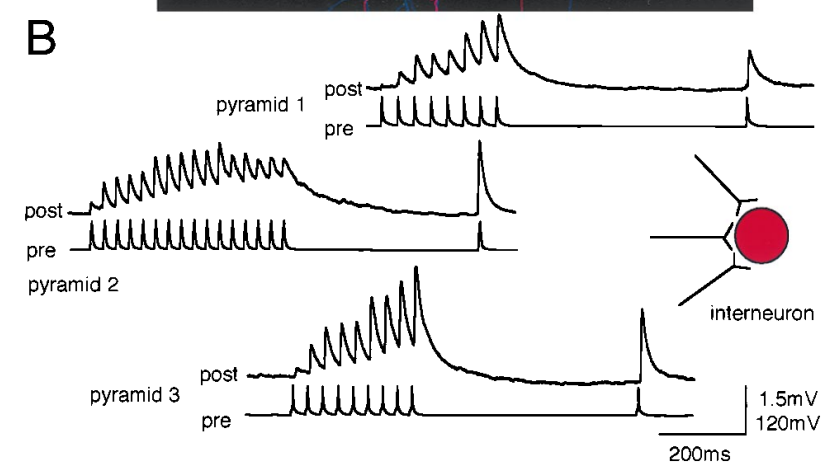


Depression

A



B



Facilitation

